

Abnormal event extraction based on large language model retrieval enhancement

Yu Zhang

Beijing Key Laboratory on Integration and Analysis of Large-scale Stream Data

College of Information, North China University of Technology

Beijing, China

zy1779691041@163.com

Abstract—In some professional fields, abnormal event extraction faces two core challenges: one is the scattering of event arguments, and the key information is often scattered in different sentences of the document, and the other is the coexistence of multiple events, which are time-related and may share device parameters. To this end, this paper proposes a document-level multi-task event extraction method DF-RAG, which deeply integrates large language models with retrieval-augmented technology to improve the ability to extract document-level abnormal events in professional fields. The method decomposes the event extraction task into two subtasks, abnormal event detection and event argument extraction, and optimizes the information extraction process through the R-Adapter network. The experimental results show that the F1 score of TP-Fault in the dataset is 75.6% by the DF-RAG method, which is significantly better than that of the baseline model, and the F1 score of the TPP-FD dataset using DF-RAG is also significantly higher than that of the baseline model. This method solves the key technical bottleneck of abnormal event extraction in the professional field, and is of great scientific significance for promoting the development of natural language processing in vertical fields.

Keywords—Event Extraction, Large Language Models, Retrieval-Augmented Technology, R-Adapter

I. INTRODUCTION

Thermal power plants generate massive volumes of operational logs during equipment operation, which record key information such as operational records and are crucial for fault prediction and maintenance decisions. However, due to the large scale and complex structure of the logs, traditional manual analysis struggles to quickly and accurately identify potential equipment anomalies. Thus, automatically extracting abnormal events from unstructured log texts has become a key challenge in improving the intelligent operation and maintenance of thermal power plants.

In practical scenarios of thermal power plants, abnormal events often involve multiple interrelated information fragments. As shown in Figure 1, taking "superheater tube burst" as an example, a complete anomaly description may include abnormal equipment parameters, alarm trigger times, and related equipment status. These key elements are usually scattered across different parts of the logs. Traditional sentence-level event extraction methods can only process information within a single sentence and cannot effectively integrate cross-sentence anomaly features. Furthermore, thermal power plant logs often contain multiple overlapping abnormal events^[1]. For instance, "superheater overtemperature" may occur simultaneously with "superheater tube burst". These events

not only have temporal correlations but may also share certain equipment parameters. Document-level event extraction methods can identify the interrelationships in such multi-event scenarios, providing a more comprehensive basis for fault diagnosis^[2]. With the breakthroughs of large language models (LLMs) in natural language processing, deep learning-based document-level event extraction technologies offer new solutions for thermal power plant log analysis, promising to significantly improve the accuracy and efficiency of abnormal event extraction^[3].

To address these challenges, this paper proposes DF-RAG, a document-level multi-task extraction method, to enhance the model's ability to extract document-level abnormal events in the thermal power domain. Its core contributions include:

- 1) Combining large language models with retrieval-augmented generation (RAG) technology to solve document-level abnormal event extraction tasks in the thermal power domain.
- 2) Introducing RAG-Adapter to effectively capture key features of retrieved information and designing retrieval-aware prompting strategies to deeply integrate retrieved information into LLM generation, enabling LLMs to better understand external documents.

This study focuses on the performance of large language models and retrieval-augmented generation technology in document-level abnormal event extraction. Experimental results show that the proposed method significantly outperforms mainstream baseline solutions.

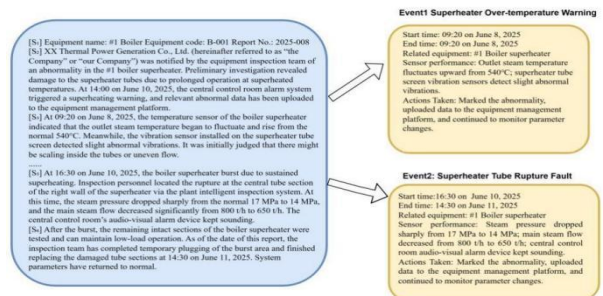


Figure 1 Example of Multi-Event Extraction.

Taking the fault reports of thermal power plants as an example, when extracting abnormal events from long documents, there exist problems of argument dispersion and the possibility of multiple events being included in a single document.

II. RELATED WORK

Event extraction (EE) aims to identify specific types of events from unstructured text and extract relevant parameters [4]. Early event extraction focused on the sentence level, with the goal of identifying events and extracting parameters from a single sentence, and could not handle the situation where event information was scattered across multiple sentences [3]. To overcome the limitations of sentence-level event extraction, document-level event extraction was later introduced, which aims to extract event information from the entire document, be able to handle cases where event parameters are scattered across multiple sentences, and model complex relationships between events [5].

With the rise of deep learning technology, convolutional neural networks (CNNs) and recurrent neural networks (RNNs) have brought new opportunities, such as Alex et al. for diabetes prediction using convolutional networks [6], Minango et al. for identifying melanoma [7], and Tataei Sarshar et al. [8] for segmenting brain tumors in MRI images using convolutional neural networks. Researchers began experimenting with deep learning models to automatically learn features and extract events. These methods typically use convolutional neural networks (CNNs), recurrent neural networks (RNNs), etc. to capture local and global information in documents, thereby improving the performance of event extraction [9]-[10]. For example, Li et al. showed that the combination of CNNs, RNNs, and DNNs is effective for relationship extraction [11]. Yu et al. proposed event extraction methods based on Tree-LSTM and Bi-GRU [12], and Alrasoud et al. also verified the effectiveness of convolutional and recurrent neural networks in sentiment analysis [13], but these methods still face challenges in dealing with complex event relationships and multi-event co-occurrence.

To overcome these challenges, researchers began exploring the use of large language models for document-level event extraction. Large language models such as the GPT series, T5, and recently various open-source LLMs have demonstrated strong capabilities in the field of natural language processing [14]-[15]. Among them, ChatGPT is a well-trained AI model that can chat almost like a human [17-18]. However, the accuracy of general-purpose large models is not high in event-specific extraction, and retrieval augmentation technology can improve the accuracy of the model's answers by retrieving relevant information from external knowledge bases [19].

III. METHOD

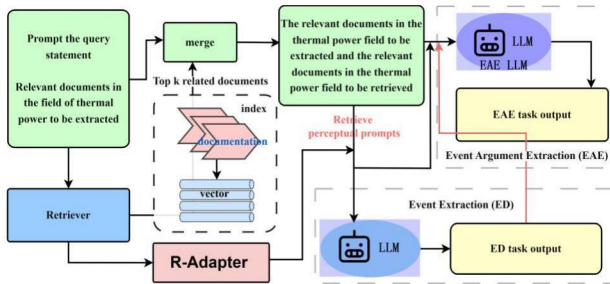


Figure 2. illustrates the model framework of DF-RAG.

Figure 2 presents the framework of the DF-RAG model. The framework decomposes document-level event extraction into two subtasks: abnormal event detection and event argument extraction. The R-Adapter network performs deep processing of retrieved documents to generate enhanced retrieval information for abnormal event detection, while transforming the retrieval results into structured features to support event argument extraction.

DF-RAG decomposes the document-level event extraction task into two sequentially solvable subtasks, namely abnormal event detection and event argument extraction, by combining large language models and retrieval-augmented generation techniques. As shown in Figure 2, the query document input to the retriever is first processed through the R-Adapter network to generate deeper retrieval information to assist the abnormal event detection task. Then, the results generated by the abnormal event detection task serve as the prerequisite for event argument extraction, and the generative representations of the queries and retrieved documents from the retriever are converted into structured features.

A. Retrieval Augmentation for Large Language Models

Traditional RAG is responsible for combining the documents returned by the retriever with a given query statement to generate prompts input to the large language model (LLM) to generate answers. Formally, given a prompt query statement q and a list of documents ranked by the retriever $\mathcal{D} = \{d_1, d_2, \dots, d_k\}$, the input to the LLM generates the output $\hat{y}$. This process can be represented as $\hat{y} = \text{LLM}(P(q, \mathcal{D}))$, where P is a predefined prompt template. This approach couples through simple prompting, leading to ineffective communication and a semantic gap.

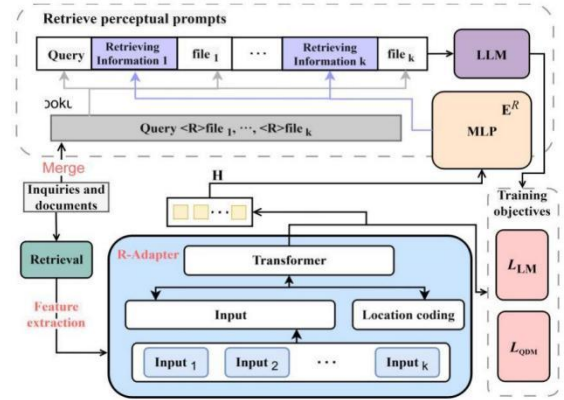


Figure 3. Retrieval Augmentation for Large Language Models.

Figure 3 shows the retrieval augmentation mechanism of DF-RAG. Unlike traditional RAG, which directly concatenates retrieved documents with queries (leading to semantic gaps), DF-RAG first transforms retriever outputs into unified features that capture query-document relationships. The R-Adapter extracts key information using self-attention to model dependencies. A retrieval-aware prompting strategy then integrates this information into the LLM's input space to enhance semantic understanding.

The DF-RAG method (Figure 3) addresses the problem through these steps: First, it converts query and document

representations from the retriever into unified features that capture fine-grained relationships between queries/documents and among documents. Next, the R-Adapter network extracts necessary retrieval information for the LLM, processing feature dependencies via self-attention. Additionally, DF-RAG uses a retrieval perception prompt strategy to embed retrieved information into the LLM's input space, avoiding information loss and complexity. This method is suitable for low-resource scenarios with a frozen LLM. High-quality retrieval features are crucial for enhancing the model's understanding of thermoelectric domain documents before anomalous event detection and feature extraction. The DF-RAG framework encodes query statements and knowledge documents into semantic vectors through a retriever. However, these single-modality representations can't be directly used for model training due to the lack of semantic interaction features needed for LLM generation. Thus, converting representations into a unified format is key to improving the model's generalization ability.

We integrate multimodal semantic representations into unified retrieval features by calculating relevance, preceding similarity, and proximity similarity scores. The specific calculation steps are as follows:

Relevance Calculation. For thermal power plant equipment fault diagnosis documents, the relevance score between the query and the i document is calculated as follows, as shown in Equation (1):

$$\begin{cases} r_i = \alpha \cdot \cos(x_i^q, x_i^d) + (1 - \alpha) \cdot \text{Jaccard}(T_q, T_d) \\ \text{Jaccard}(T_q, T_d) = |T_q \cap T_d| / |T_q \cup T_d| \end{cases} \quad (1)$$

where $\alpha \in [0, 1]$ is the weight coefficient (default: 0.5), $\text{Jaccard}(T_q, T_d)$ calculates the Jaccard similarity between query terms and document terms (particularly suitable for matching professional terminology in thermal power plant equipment fault descriptions), T_q represents the term set in the prompt query text, and T_d represents the term set in the i thermal power domain knowledge document.

Preceding Similarity Calculation. This calculates the similarity score between the representation of the i thermal power domain knowledge document and its preceding weighted representation, as shown in Equation (2):

$$\gamma_i = \beta \cdot \text{sim}(x_i^d, x_{i-1}^d) + (1 - \beta) \cdot \text{DTW}(s_i, s_{i-1}) \quad (2)$$

where the $\text{DTW}(\cdot)$ function analyzes the similarity of fault time-series (particularly suitable for temporal analysis of fault progression in thermal power plants), $\beta \in [0, 1]$ is the weight coefficient, s_i represents the operational time-series data of the equipment corresponding to the i document, and s_{i-1} represents the time-series data corresponding to the preceding document.

Proximity Similarity Calculation. This calculates the average similarity between the i thermal power domain knowledge document and its adjacent documents. Specifically, the proximity similarity of the i document in the ranked list is defined as, as shown in Equation (3):

$$\zeta_i = \frac{1}{2} [\text{sim}_G(x_i^d, x_{i-1}^d) + \text{sim}_G(x_i^d, x_{i+1}^d)] \quad (3)$$

where $\text{sim}_G(\cdot)$ uses a graph similarity algorithm (considering topological relationships between thermal power plant equipment systems, particularly suitable for multi-system correlated fault analysis in thermal power plants), and ζ_i represents the average similarity between the i document and its adjacent documents.

Finally, these three features are concatenated to generate input_i as the input for the i document. The feature list $\{\text{input}_i\}_{i=1}^k$ is then fed into the R-Adapter network to further leverage the retrieval information, as shown in Equation (4).

$$\text{input}_i = \{r_i, \gamma_i, \zeta_i\} \quad (4)$$

R-Adapter Network. Inspired by Li et al., we designed the R-Adapter specifically for thermal power plants' characteristics. Documents in thermal power plants are multimodal, highly specialized, and system-coupled, making it hard for traditional methods to capture these features fully. The modified R-Adapter bridges domain-specific retrievers and LLMs, filling the semantic gap. As shown at the bottom of Figure 3, the R-Adapter is a pluggable Transformer-based model that takes list-level features as input and outputs retrieval information. To better understand these features, an input embedding layer linearly transforms them into a higher-dimensional space, with positional embeddings added before attention encoding to maintain sequence awareness. Then, the Transformer encoder processes the input sequence using self-attention masks. For an input list, the process is expressed in Equation (5):

$$H = f_{\text{att}}[f_{\rightarrow h_1}(\{\text{input}_i\}_{i=1}^k) + p] \quad (5)$$

where f_{att} is a Transformer encoder with h_1 hidden dimensions, $f_{\rightarrow h_1}$ denotes a linear mapping layer, p represents trainable positional embeddings, and H is the output embedding containing deeper retrieval information, ultimately passed to the LLM for answer generation.

Retrieval-Aware Prompting. During the generation process, it is crucial for the LLM to effectively utilize retrieved thermal power plant document information. As shown in the upper part of Figure 3, we introduce a retrieval-aware prompting strategy that injects retrieval information extracted by the R-Adapter into the LLM's generation process. First, we use a projection layer to linearly transform the retrieval information into the same dimension as the LLM's token embedding layer. Formally, this is expressed as shown in Equation (6):

$$E^R = f_{\rightarrow h_2}(H) = \text{LayerNorm}(W_p H + b_p) \quad (6)$$

Where $f_{\rightarrow h_2}$ is a linear projection layer through an MLP, h_2 is the dimension of the LLM's token embedding layer, and $W_p \in \mathbb{R}^{h_2 \times h_1}$ is a projection matrix adapted for the thermal power domain. LayerNorm is specifically added to handle numerical fluctuations in thermal power plant documents. Next, we tokenize the query and documents using the LLM's tokenizer and convert them into embeddings. For

example, document d is tokenized into $t_d = \{t_j\}_{j=1}^{n_j}$, where t_d is the j token in document, and n_j is the number of tokens in document d . The token embeddings are then converted via the token embedding lookup layer, as shown in Equation (7):

$$E^d = \text{Concat}[f_{\text{emb}}(t_d), M_{\text{term}}] \quad (7)$$

where f_{emb} is the LLM's token embedding layer, $E_d \in \mathbb{R}^{n_d \times h_2}$ is the embedding of document d , and M_{term} is a thermal power terminology embedding matrix containing 3000+ specialized terms. Similarly, the query embedding $E_q = \{e_j\}_{j=1}^{n_j}$ can be obtained, where n_j is the number of query tokens.

To enable fine-grained analysis of each document, the corresponding retrieval information embedding E^R is prepended to each document embedding. These serve as external knowledge anchors, guiding the LLM to focus on useful documents. The final input embeddings are arranged as shown in Equation (8):

$$E = \begin{bmatrix} E_q & E_{\text{sys}} \\ E_1^R & E_1^d \\ \vdots & \vdots \\ E_k^R & E_k^d \end{bmatrix} \quad (8)$$

where E_{sys} denotes equipment system topology encoding (e.g., boiler, steam turbine, generator), and E^R is the retrieval information embedding for the i document. This approach effectively integrates retrieval information with corresponding documents, reducing the LLM's burden in processing all documents. Finally, the model's output response $\hat{y}$ is obtained via $\hat{y} = \text{LLM}(E)$.

B. Document Event Extraction

Event Detection Task. Based on the retrieved thermal power plant documents most similar to the target document, the event type and event detection task instructions are combined with the content of the document to be detected and input to the LLM to obtain event detection results.

Event Argument Extraction (EAE) follows event detection. Predefined event schemas identify argument roles for each event type. These roles are combined with task-specific instructions to create EAE prompts for the LLM. The model is fine-tuned with supervised data labeled with event arguments to improve accuracy in extracting relevant arguments.

IV. EXPERIMENT

A. Dataset

TABLE 1 DATASET DISTRIBUTION

	Training Set	Validation Set	Test Set
TP-Fault	660	220	220
TPP-FD	480	160	160
Total	1140	380	380

This paper conducts experimental evaluations on two document-level datasets for electrical equipment anomaly events: the Thermal Power Fault dataset (TP-Fault) and the Thermal Power Plant Fault dataset (TPP-FD), with specific data distributions shown in Table 1.

The TP-Fault is a Chinese document-level event extraction dataset for thermal power equipment failures, sourced from crawled fault records of thermal power plants, containing a total of 1,100 samples. The TPP-FD dataset originates from a collaboration between universities and power plants, comprising 800 desensitized records of boiler and steam turbine anomalies (2018-2022). Both datasets were partitioned into training, validation, and test sets using a 6:2:2 ratio.

B. Baseline

In order to evaluate the performance of the DF-RAG method, the following baseline models are selected for comparison, which cover the methods of direct event extraction with large language models to extraction with retrieval enhancement technology in combination with the knowledge base to ensure the comprehensiveness and fairness of the comparison. The large language models selected are DeepSeek-R1 and Qwen2.5-3B.

C. Experimental Configuration

The temperature is set to 0.1, the rank of the LoRA adapter is 8, the alpha value is 16, the batch size per GPU is 2, the gradient cumulative step is 8, and the learning rate is $5e-5$. The experiment was conducted on NVIDIA GeForce RTX4090 (24GB) GPUs. In terms of evaluation indicators, we used the system from previous studies. Predictive records are chosen by matching the same event type and sharing the most with standard records. F1 scores are then calculated based on parameter comparisons to provide a comprehensive and objective assessment of the model's performance in document-level event extraction.

D. Comparative Experiments

TABLE 2 ARGUMENT EXTRACTION F1 SCORES ON TP-FAULT AND TPP-FD DATASETS

Model	knowledge base	TP-Fault	TPP-FD
DeepSeek-R1	×	70.2	53.6
Qwen2.5-3B	×	62.5	43.7
DeepSeek-R1	✓	71.7	54.1
Qwen2.5-3B	✓	63.1	51.9
DF-RAG	✓	75.6	59.7

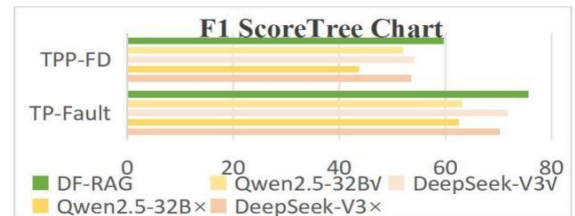


Figure 4. Bar Chart of Argument Extraction F1 Scores on the TP-Fault and TPP-FD Datasets.

Figure 4 shows the comparison of F1 scores between the DF-RAG method and baseline models in the event argument extraction task on the TP-Fault and TPP-FD datasets.

In this paper, the performance of the DF-RAG method is evaluated through comparative experiments, as shown in Table 2 and Figure 4. The results show that DF-RAG achieves an F1 score of 75.6% on the TP-Fault dataset, significantly outperforming baseline models like DeepSeek-R1 (70.2% without KB, 71.7% with KB) and Qwen2.5-3B (62.5% without KB, 63.1% with KB). On the TPP-FD dataset, DF-RAG's F1 score is 59.7%, also higher than the baselines. This demonstrates DF-RAG's superiority in document-level event extraction for thermopower, effectively handling complex scenarios with scattered elements and multiple intertwined events, and achieving deep retrieval-generation collaboration.

V. CONCLUSION

The DF-RAG method proposed in this paper combines LLM and RAG technologies to decompose document-level event extraction into two subtasks: abnormal event detection and event argument extraction. It also designs the R-Adapter network and retrieval-aware prompting strategies, which significantly enhance the model's semantic understanding and event extraction capabilities. Experiments show that DF-RAG achieves higher F1 scores in event argument extraction on the TP-Fault and TPP-FD datasets compared to mainstream baseline models. This demonstrates its effectiveness in handling scenarios with cross-sentence argument dispersion and multi-event interweaving, providing strong support for intelligent operation and maintenance in thermal power plants.

VI. EXPECTATION

In the future, the DF-RAG method can be optimized from three aspects: first, expand the scale and type of datasets and enhance the generalization ability; second, combine real-time data flow processing to realize real-time monitoring of operating status and dynamic event extraction; The third is to explore the integration with cutting-edge technology to improve performance. These improvements will make DF-RAG more valuable in the field of industrial intelligent operation and maintenance, providing direction for future research.

REFERENCES

- [1] Xu, Y., Feng, C., Wang, B., Huang, J., & Qi, X. S. (2023). Retrieval-Augmented Document-Level Event Extraction with Cross-Attention Fusion. In *Communications in computer and information science*.
- [2] Qin, G., Lin, N., Shen, M., Bai, Q., Zhou, D., & Yang, A. (2024). Global information enhancement and subgraph-level weakly contrastive learning for lightweight weakly supervised document-level event extraction. *Expert Systems with Applications*, 240, 122516. <https://doi.org/10.1016/j.eswa.2023.122516>
- [3] Chu, T., Chen, Y., Wilson, J. P., & Liu, L. (2025). A large language model-enhanced argument extraction and clustering model for urban hotspot event detection using crowdsourced data. *Expert Systems with Applications*, 293, 128760. <https://doi.org/10.1016/j.eswa.2025.128760>
- [4] Hatwell, J., Gaber, M. M., & Azad, R. M. A. (2020). CHIRPS: Explaining random forest classification. *Artificial Intelligence Review*, 53(8), 5747–5788. <https://doi.org/10.1007/s10462-020-09833-6>
- [5] Tong, M., Xu, B., Wang, S., Han, M., Cao, Y., Zhu, J., Chen, S., Hou, L., & Li, J. (2022). DocEE: A Large-Scale and Fine-grained Benchmark for Document-level Event Extraction. *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. <https://doi.org/10.18653/v1/2022.naacl-main.291>
- [6] Alex, S. A., de Oliveira, G. G., Iano, Y. (2023). Pregnant Women Diabetic Prediction Using 1D-Convolutional Neural Network and SMOTE Procedure. In: Iano, Y., Saotome, O., Kemper Vásquez, G. L., de Moraes Gomes Rosa, M. T., Arthur, R., Gomes de Oliveira, G. (eds) *Proceedings of the 8th Brazilian Technology Symposium (BTSym, 22)*. BTSym 2022. Smart Innovation, Systems and Technologies, vol 353. Springer, Cham. https://doi.org/10.1007/978-3-031-31007-2_29
- [7] Minango, P. et al. (2021). Melanoma Classification Based on Three Different Very Deep Neural Networks. In: Iano, Y., Saotome, O., Kemper, G., Mendes de Seixas, A. C., Gomes de Oliveira, G. (eds) *Proceedings of the 6th Brazilian Technology Symposium (BTSym, 20)*. BTSym 2020. Smart Innovation, Systems and Technologies, vol 233. Springer, Cham. https://doi.org/10.1007/978-3-030-75680-2_52
- [8] Tataei Sarshar, N. et al. (2023). Glioma Brain Tumor Segmentation in Four MRI Modalities Using a Convolutional Neural Network and Based on a Transfer Learning Method. In: Iano, Y., Saotome, O., Kemper Vásquez, G. L., Cotrim Pezzuto, C., Arthur, R., Gomes de Oliveira, G. (eds) *Proceedings of the 7th Brazilian Technology Symposium (BTSym, 21)*. BT Sym 2021. Smart Innovation, Systems and Technologies, vol 207. Springer, Cham. https://doi.org/10.1007/978-3-031-04435-9_39
- [9] Yang, Y., Wu, Z., Yang, Y., Lian, S., Guo, F., & Wang, Z. (2022). A Survey of Information Extraction Based on Deep Learning. *Applied Sciences*, 12(19), 9691. <https://doi.org/10.3390/app12199691>
- [10] Ku, A. Y., & Hool, A. (2024). Capabilities and limitations of AI Large Language Models (LLMs) for materials criticality research. *Mineral Economics*. <https://doi.org/10.1007/s13563-024-00478-3>
- [11] Birur, N. A., Baswa, T., Kumar, D., Loya, J., Agarwal, S., & Harshangi, P. (2024). VERA: Validation and Enhancement for Retrieval Augmented systems (Version 1). *arXiv*. <https://doi.org/10.48550/ARXIV.2409.15364>
- [12] Usama, M., Ahmad, B., Song, E., Hossain, M. S., Alrashoud, M., & Muhammad, G. (2020). Attention-based sentiment analysis using convolutional and recurrent neural network. *Future Generation Computer Systems*, 113, 571–578. <https://doi.org/10.1016/j.future.2020.07.022>
- [13] Shiri, F., Nguyen, V., Moghimifar, F., Yoo, J., Haffari, G., & Li, Y.-F. (2024). Decompose, Enrich, and Extract! Schema-aware Event Extraction using LLMs.
- [14] Vagliano, I., Dormosh, N., Rios, M., Luik, T. T., Buonocore, T. M., Elbers, P. W. G., Dongelmans, D. A., Schut, M. C., & Abu-Hanna, A. (2023). Prognostic models of in-hospital mortality of intensive care patients using neural representation of unstructured text: A systematic review and critical appraisal. *Journal of Biomedical Informatics*, 146, 104504. <https://doi.org/10.1016/j.jbi.2023.104504>
- [15] Cai, E., & O'Connor, B. (2023). A Monte Carlo Language Model Pipeline for Zero-Shot Sociopolitical Event Extraction (Version 2). *arXiv*. <https://doi.org/10.48550/ARXIV.2305.15051>
- [16] Keith Norambuena, B. F., Mitra, T., & North, C. (2023). A Survey on Event-Based News Narrative Extraction. *ACM Computing Surveys*, 55(14s), 1–39. <https://doi.org/10.1145/3584741>
- [17] Chuma, E. L., & de Oliveira, G. G. (2023). Generative AI for Business Decision-Making: A Case of ChatGPT. *Management Science and Business Decisions*, 3(1), 5–11. <https://doi.org/10.5281/msbd.63>
- [18] Chuma, E. L., Alves, A. M., & de Oliveira, G. G. (2024). Evolution of Generative AI for Business Decision-Making: A Case of ChatGPT. *Management Science and Business Decisions*, 4(1), 5–14. <https://doi.org/10.5281/msbd.87>
- [19] Kachuee, M., Ahuja, S., Kumar, V., Xu, P., & Liu, X. (2024). Improving Tool Retrieval by Leveraging Large Language Models for Query Generation. *arXiv*. <https://doi.org/10.48550/ARXIV.2412.03573>